

Safety data sheet

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BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 06.07.2021

Version: 1.0

Product: **GARDOCLEAN S 5225**

(ID no. 30690842/SDS_GEN_00/EN)

Date of print 11.03.2022

1. Identification

Product identifier

GARDOCLEAN S 5225

Recommended use: Pre-treatment of metal surfaces.

Not recommended use: Uses other than recommended

Details of the supplier of the safety data sheet

Company:
Chemetall GmbH
Trakehner Straße, 3
60487, Frankfurt am Main
Germany
+49(0)69 7165-0
sds.global-chemetall@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (oral)
Skin Corr./Irrit. 1A
Eye Dam./Irrit. 1

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For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System, UN (GHS)

Pictogram:



Signal Word:

Danger

Hazard Statement:

H302

Harmful if swallowed.

H314

Causes severe skin burns and eye damage.

Precautionary Statements (Prevention):

P280

Wear protective gloves, protective clothing and eye protection or face protection.

P264

Wash contaminated body parts thoroughly after handling.

P270

Do not eat, drink or smoke when using this product.

P260

Do not breathe dust or mist.

Precautionary Statements (Response):

P305 + P351 + P338

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P304 + P340

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P303 + P361 + P353

IF ON SKIN (or hair): Remove or Take off immediately all contaminated clothing. Rinse skin with water or shower.

P363

Wash contaminated clothing before reuse.

P330

Rinse mouth

P310

Immediately call a POISON CENTER or physician.

P301 + P330 + P331

IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

Precautionary Statements (Storage):

P405

Store locked up.

Precautionary Statements (Disposal):

P501

Dispose of contents and container to hazardous or special waste collection point.

Potassium hydroxide

Other hazards

According to UN GHS criteria

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If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Water, inorganic compounds, inorganic base

Hazardous ingredients (GHS)

According to UN GHS criteria

Potassium hydroxide

Content (W/W): $\geq 30\%$ - $< 50\%$

CAS Number: 1310-58-3

EC-Number: 215-181-3

INDEX-Number: 019-002-00-8

Met. Corr. 1

Acute Tox. 4 (oral)

Skin Corr./Irrit. 1A

Eye Dam./Irrit. 1

H290, H302, H314

Specific concentration limit:

Skin Corr./Irrit. 2: $0,5\%$ - $< 2\%$

Eye Dam./Irrit. 2: $0,5\%$ - $< 2\%$

Skin Corr./Irrit. 1A: $\geq 5\%$

Skin Corr./Irrit. 1B: 2% - $< 5\%$

Potassium silicate

Content (W/W): $\geq 5\%$ - $< 7\%$

CAS Number: 1312-76-1

EC-Number: 215-199-1

REACH registration number: 01-2119456888-17

Skin Corr./Irrit. 2

Eye Dam./Irrit. 1

H318, H315

Tetrapotassium pyrophosphate

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Content (W/W): $\geq 10\%$ - $< 12,5\%$ Eye Dam./Irrit. 2A
H319
CAS Number: 7320-34-5
EC-Number: 230-785-7
REACH registration number: 01-
2119489369-18

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

If inhaled:

Remove the affected individual into fresh air and keep the person calm. If breathing is irregular or stopped, administer artificial respiration.

On skin contact:

Flush with copious amounts of water for at least 15 minutes. Immediate medical attention required.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist. Immediate medical attention required. Remove contact lenses, if present.

On ingestion:

Rinse mouth immediately with water. Do not give solids or liquids. Do not induce vomiting. Immediate medical attention required.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Hazards: May cause severe burns of the mouth and throat if orally ingested, as well as a danger of perforation of the oesophagus and the stomach.

Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:

carbon dioxide, dry powder, alcohol-resistant foam, water spray

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Hazardous decomposition products formed under fire conditions.

Advice for fire-fighters

Special protective equipment:
Appropriate breathing apparatus may be required.

Further information:

Cool closed containers in the vicinity of the source of fire. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

For non-emergency personnel: Use personal protective clothing. Ensure adequate ventilation. Keep away from sources of ignition. For emergency responders: Advice on product handling can be found in sections 7 and 8 of this safety data sheet. Information regarding personal protective measures, see section 8.

Environmental precautions

Do not allow to enter drains or waterways. Do not discharge into the subsoil/soil. If the product enters drains or sewers, the local water company should be contacted immediately; in the case of contamination of streams, rivers or lakes, the Environment Agency.

Methods and material for containment and cleaning up

Ensure adequate ventilation. Pick up with inert absorbent material (e.g. sand, earth etc.). Take up mechanically and collect in suitable container (adequately labelled) for disposal.

7. Handling and Storage

Precautions for safe handling

Provide good ventilation of working area (local exhaust ventilation if necessary). Do not return residues to the storage containers. Smoking, eating and drinking are forbidden in application area. For personal protection see section 8. Comply with the health and safety at work laws. The workplace should be equipped with an emergency shower and eye-rinsing facility. Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Avoid all sources of ignition: heat, sparks, open flame. The relevant fire protection measures should be noted.

Conditions for safe storage, including any incompatibilities

Segregate from acids.

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Polyethyleneterephthalate (PET), Polypropylene (PP), Carbon steel (Iron), tinned carbon steel (Tinplate)

Further information on storage conditions: Avoid contact with amphoteric metals (e.g. aluminium, lead, zinc). Protect from frost.

The entrance to storage rooms is to be granted only to appropriately trained personnel. Keep only in the original container. Keep container dry. Keep in a cool, well-ventilated place. Keep away from heat. Avoid direct sunlight.

Storage stability:

Storage temperature: -14 - 70 °C

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection**Control parameters**Components with occupational exposure limits

No occupational exposure limits known.

Exposure controlsPersonal protective equipment

Respiratory protection:

Respiratory protection required if exposure limit (if available) may be exceeded (Gas filter EN 14387 B)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

butyl rubber gloves - material thickness: 0.5 mm

nitrile rubber (NBR) - 0.4 mm coating thickness

Performance level 6, corresponding to a breakthrough time of >480 min according to EN ISO 374-1

The protection glove should be tested for its specific suitability (e.g. mechanical strength, product compatibility, anti-static properties).

The gloves should be replaced immediately in case of damage or signs of wear. It is recommended to use preventative skin protection (skin cream).

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

Chemical resistant protective clothing according to DIN EN 13034 (Type 6)

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General safety and hygiene measures

Eye wash fountains and safety showers must be easily accessible. Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation. This can be achieved by the use of local exhaust ventilation and good general extraction. Remove contaminated clothing immediately and dispose of safely. Hands and/or face should be washed before breaks and at the end of the shift. Keep separated from food stuffs and feed stocks.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	colourless to yellowish
pH value:	14,0
Melting point:	
	not determined
onset of boiling:	
	not determined
Flash point:	
	not applicable
Flammability:	not applicable
Lower explosion limit:	
	not determined
Ignition temperature:	
	not determined
Vapour pressure:	
	(20 °C)
	not determined
	(50 °C)
	not determined
Density:	1,548 g/cm ³
	(20 °C)
Viscosity, kinematic:	10,0 mm ² /s
	(20 °C)
	(40 °C)
	not determined
Explosion hazard:	not explosive
Fire promoting properties:	not fire-propagating

Other information

Self heating ability:	It is not a substance capable of spontaneous heating.
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Miscibility with water:

No data available.

Flow time:

35 s

(DIN EN ISO 2431; 3 mm)

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effect on metals.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

May react with aluminum, zinc or tin, with formation of hydrogen.

May react with aluminum, zinc or tin, with formation of hydrogen.

Conditions to avoid

Avoid direct sunlight.

Incompatible materials

Substances to avoid:

acids, light metals

Hazardous decomposition products

:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion.

*Information on: Potassium hydroxide**Experimental/calculated data:**LD50 rat (oral): 333 mg/kg (OECD Guideline 425)**Literature data.*

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Irritation

Assessment of irritating effects:

Highly corrosive! Damages skin and eyes. May cause severe damage to the eyes.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:

Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

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12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

There are no test results available for this product. Do not allow to enter drains or waterways. Based on available data, the classification criteria are not met.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

No data available concerning biodegradation and elimination.

Bioaccumulative potential

Bioaccumulation potential:

No data available.

Mobility in soil

Assessment transport between environmental compartments:

Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:

Do not allow to enter drains or waterways. Do not allow to enter soil, waterways or waste water channels.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

No disposal via sewage or waste water systems.

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Contaminated packaging:

Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

UN number	UN1814
UN proper shipping name:	POTASSIUM HYDROXIDE SOLUTION
Transport hazard class(es):	8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	Tunnel code: E

RID

UN number	UN1814
UN proper shipping name:	POTASSIUM HYDROXIDE SOLUTION
Transport hazard class(es):	8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	None known

Inland waterway transport

ADN

UN number	UN1814
UN proper shipping name:	POTASSIUM HYDROXIDE SOLUTION
Transport hazard class(es):	8
Packing group:	II
Environmental hazards:	no
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

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IMDG

UN number:	UN 1814
UN proper shipping name:	POTASSIUM HYDROXIDE SOLUTION
Transport hazard class(es):	8
Packing group:	II
Environmental hazards:	no
	Marine pollutant: NO
Special precautions for user:	None known

Air transport**IATA/ICAO**

UN number:	UN 1814
UN proper shipping name:	POTASSIUM HYDROXIDE SOLUTION
Transport hazard class(es):	8
Packing group:	II
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Met. Corr.	Corrosive to metals
H290	May be corrosive to metals.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.

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H318	Causes serious eye damage.
H315	Causes skin irritation.
H319	Causes serious eye irritation.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.